MIL-A-12636B

24 January 1969

SUPERSEDING
MIL-T-12636A
23 March 1959

MILITARY SPECIFICATION

ANALYZER ZM-3( )/U

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers one type of test set, designated Analyzer ZM-3( )/U and Case, Test Set CY-1121( )/U, for measuring the essential characteristics of capacitors.

2. APPLICABLE DOCUMENTS.

2.1 Documents. The following documents of the issue in effect at date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton, Wrapping.
PPP-B-636	Box, Fiberboard.
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-291	Paperboard, Wrapping, Cushioning.
PPP-S-760	Strapping, Nonmetallic, (and Connectors).
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

FSC 6625

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MIL-E-75	Electron, Tube Packing, Packaging, and Container Marking for General Specification for.
MIL-P-116	Preservation, Method of.
MIL-P-11268	Parts, Materials and Processes Used in Electronic Communication Equipment.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-B-43291	Boxes; Fiberboard; Corrugated, Double-Wall, Weather Resistant.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Pallatized Unit Loads; 40 in. x 48 in. 4-Way Partial and 4-Way Pallets.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-781	Reliability Tests Exponential Distribution.
MIL-STD-810	Military Standard Environmental Test Methods for Aerospace and Ground Equipment.

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package.
SC-DL-51826	Analyzer ZM-3( )/U.
SC-DL-90672	Case, Test Set, CY-1121( )/U.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

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3. REQUIREMENTS.

3.1 Description. The Analyzer ZM-3( )/U is used to determine characteristics of capacitors from 5 picofarads (pF) to 10,000 microfarads (uF). The analyzer is used to measure leakage current of electrolytic capacitors, insulation resistance of all types of capacitors except electrolytic capacitors, capacitance of all types of capacitors, and the power factor of capacitors larger than .04 uF. The analyzer may also be used to check insulation resistance and capacitance of cables. The functions and ranges shall be as follows:

a. Capacitance ranges - 5 to 100 pF, 80 to 50,000 pF, .04 to 30 uF, 25 to 1,000 uF, and 250 to 10,000 uF.

b. Insulation resistance ranges - 1.0 to 100 megohms, and 110 to 10,000 megohms.

c. Power factor ranges - 0 to 50 on capacitors of 0.05 to 10,000 uF.

d. Leakage - 0 to 60 Vdc, and 0 to 600 Vdc.

e. Leakage current range - 0 to 50 mA dc.

3.2 First article samples. The contractor shall furnish four (4) complete first article samples for approval provided the invitation for bid and the contract requires first article samples.

3.3 Construction. The equipment shall be constructed in accordance with the following drawings and the requirements specified herein:

SC-DL-51826	Analyzer ZM-3( )/U
SC-DL-90672	Case, Test Set, CY-1121( )/U

3.4 Parts, materials and processes-general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection of requirements therein, shall apply.

3.5 Finish, protective. The equipment shall be finished in accordance with MIL-F-14072. This includes finish of hardware; such as screws, etc., and necessary touch-up after mounting.

3.6 Marking. Marking shall conform to MIL-M-13231.

3.6.1 Visibility. Wherever practicable, parts and assemblies shall be so mounted that their identification markings shall be readily visible with a minimum disassembly of the equipment.

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3.6.2 Serial numbers. Serial numbers are required on each nomenclatured and type designated item, except cords and cables.

3.7 Electrical.

3.7.1 Accuracy. The accuracy of the analyzer in all functions shall be within tolerances specified.

3.7.1.1 Capacitance. The accuracy of the capacitance measurements shall be as follows, when tested in accordance with 4.7.1.1:

a. 5 pF to 100 pF - ±3 percent

b. 101 pF to 100 uF - ±5 percent

c. 101 uF to 1,000 uF - ±10 percent

d. 1001 uF to 10,000 uF - ±20 percent

3.7.1.2 Insulation resistance. The insulation resistance indicated by the analyzer shall be within ±20 percent of resistance measured, when tested in accordance with 4.7.1.2.

3.7.1.3 Power factor. The power factor shall be within ±10 percent of the power indication when tested in accordance with 4.7.1.3.

3.7.1.4 Leakage. The leakage current indicated by the analyzer shall be within 3 percent of full scale meter deflection at any part of the scale. The voltage indicated by the analyzer shall be within 3 percent of full scale meter deflection at any part of the scale. The analyzer shall be tested in accordance with 4.7.1.4.

3.7.2 Voltage on capacitance terminals. The maximum ac voltage appearing across the capacitance test terminals, with the capacitor under test across the terminals, shall not exceed the following at any portion of its range of measurements, when tested in accordance with 4.7.2:

a. Range 2 (5 to 100 pF); 50 Vac

b. Range 3 (80 to 50,000 pF); 20 Vac

c. Range 4 (.04 to 30 uF); 10 Vac

d. Range 5 (25 to 10,000 uF); 7 Vac

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3.7.3 Discharging circuits. No charge shall remain on any capacitor when it is removed from the leakage terminals of the analyzer, after the capacitor has been tested. The analyzer shall be tested in accordance with 4.7.3.

3.7.4 Overload. Shorted capacitors applied at the leakage, insulation resistance, or capacitance terminals shall not cause damage to any components in the analyzer when left on continuously for 5 minutes and shall meet the requirements of 3.7.1.1, when tested in accordance with 4.7.4.

3.7.5 Drum dial control. The drum dial control shall be capable of 300 operations and shall meet the requirements of 3.7.1.1 when tested in accordance with 4.7.5.

3.7.6 Input power. The capacitance reading on the analyzer with a standard 50 pF, ±1 percent capacitor attached to the capacitance terminals shall not vary over ±3 pF, when the ac input voltage is varied from 97.75 to 132.25 and 195.5 to 264.5, at 50, 60, 400 and 1,000 Hz, when tested in accordance with 4.7.6.

3.7.7 Operational. Each analyzer shall be checked at the middle of each range and meet the applicable requirements of 3.7.1 when tested in accordance with 4.7.7.

3.8 Service conditions. The Analyzer ZM-3( )/U shall be constructed so that no fixed part shall become loose, cracked, flaked, pitted or blistered, no movable part or control shall be shifted in setting, position or adjustment, and no permanent degradation shall be caused in performance below that specified herein when subjected to the following conditions.

3.8.1 Fungus. The analyzer shall show no evidence of viable fungus and shall meet the requirements of 3.7.7 when tested in accordance with 4.8.1.

3.8.2 Salt fog. The analyzer shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface, and shall meet the requirements of 3.7.7 when tested in accordance with 4.8.2.

3.8.3 Temperature. The analyzer shall be subjected to temperature extremes specified.

3.8.3.1 Low temperature. The analyzer shall be subjected to the low temperature extremes of -4°F operating and -65°F storage and shall meet the electrical requirements of 3.7.7 at times specified when tested in accordance with 4.8.3.1.

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3.8.3.2 High temperature. The analyzer shall be subjected to the high temperature extremes of 125°F operating and 160°F storage, and shall meet the electrical requirements of 3.7.7 at times specified when tested in accordance with 4.8.3.2.

3.8.4 Humidity. The analyzer shall exhibit no physical damage of parts and material, such as corrosion, rust, blistering, swelling, deterioration, etc., or electrical degradation, and shall meet the electrical requirements of 3.7.7, when tested in accordance with 4.8.4.

3.8.5 Shock, drop. The analyzer shall show no evidence of physical damage except minor surface abrasion and shall meet the electrical requirements of 3.7.7 when tested in accordance with 4.8.5.

3.8.6 Altitude. The analyzer shall be subjected to the altitudes specified and shall meet the electrical requirements of 3.7.7 at times specified when tested in accordance with 4.8.6.

3.8.7 Immersion. The case CY-1121( )/U shall show no evidence of leakage when tested in accordance with 4.8.7.

3.8.8 Vibration. The analyzer shall show no evidence of breakage or loosening of parts, and shall meet the electrical requirements of 3.7.7 when tested in accordance with 4.8.8.

3.8.9 Bounce. The analyzer shall show no evidence of physical damage except minor surface abrasion and shall meet the electrical requirements of 3.7.7 when tested in accordance with 4.8.9.

3.8.10 Shock, bench-handling. The analyzer shall show no evidence of physical damage except minor surface abrasion and shall meet the electrical requirements of 3.7.7 when tested in accordance with 4.8.10.

3.9 Visual and mechanical. The analyzer shall meet the requirements of MIL-STD-252 as specified in 4.9.

3.10 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be hand picked for fit or performance. Reliance shall not be placed on any unspecified dimensions, rating, characteristics, etc. Where selection of a component having special characteristics is required, such selection shall be made from qualified or standard parts where possible and the specific characteristics of the selected part shall be defined (see 4.10).

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3.11 Burn-in and debugging. Each analyzer produced shall be subjected to the burn-in and debugging test of 4.11 after bounce preconditioning and prior to Group A inspection. The analyzer shall be monitored for operational requirement and test of 3.7.7 and 4.7.7 during period of test.

3.12 Bounce preconditioning. Each analyzer produced shall be subjected to the bounce preconditioning test of 4.6 prior to Group A inspection. After bounce preconditioning, the analyzer shall not be repaired, aligned, cleaned or otherwise changed prior to performing Group A inspection.

3.13 Workmanship. The analyzer shall be manufactured and assembled in accordance with the applicable portions of the following:

(a) In MIL-P-11268:

General requirements for plastic material and parts.
Wiring and cabling including:

Slack.
Protection.
Clearance.
Splicing.
Connectors, general.
Grounding, general.

Flux and cleaning agents for soldering.
Process for soldering.
Cleaning of equipment
General process for securing of parts.
Process of welding.
Connectors, general.
Mechanical operation of controls.
Tropicalization of material.
Lubricants.

(b) In this specification

Marking.
Finish.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the contract and specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection (Does not include preparation for delivery) (see 4.3)

(b) Inspection covered by subsidiary documents (see 4.4).

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.12).

4.3 First article inspection. This inspection shall be performed by the contractor and witnessed by a Government representative unless otherwise specified in the contract. It shall consist of Group A, B and C inspections (see Tables I, II and III respectively) on the four (4) samples of the completely assembled Analyzer ZM-3( )/U. The first article samples shall be inspected in this order: Group A and B for all units; and designated units for specific Group C inspections. Designation of units to specific Group C inspections shall be determined by the Government production engineer. After completion of all Group C tests, conforming units must be reinspected and pass all Group A inspections.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.5
Marking	3.6
Workmanship	3.13

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

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4.5.1 Group A inspection. Each unit produced shall be inspected by the contractor for conformance to all the Group A requirements and tests in Table I. Each lot shall be subject to Government verification and this inspection, including sampling, shall conform to Table I and the ordinary inspection procedures of MIL-STD-105.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in any order which is satisfactory to the Government.

TABLE I. Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and mechanical	3.9	4.9	1 dphu*	4 dphu
Electrical				
Accuracy	3.7.1	4.7.1)		
Capacitance	3.7.1.1	4.7.1.1)		
Insulation resistance	3.7.1.2	4.7.1.2)	1.5 dphu	**
Power factor	3.7.1.3	4.7.1.3)	for the	
Leakage	3.7.1.4	4.7.1.4)	group	
		)		
Voltage on capacitance		)		
terminals	3.7.2	4.7.2)		
Discharging circuits	3.7.3	4.7.3)		
Operational	3.7.7	4.7.7)	0 dphu	**

* Defects per hundred units.
** All electrical tests are major.

4.5.2 Group B inspection. This inspection, including sampling, shall conform to Table II and to the special procedures for small sample inspection of MIL-STD-105. Group B inspection shall be performed on inspection lots that have passed Group A inspection, and on samples selected from units that have been subjected to and meet the Group A inspection.

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4.5.2.1 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

TABLE II. Group B inspection

Inspection	Req. Para.	Test Para.	AQL
Interchangeability	3.10	4.10	4 dphu *
Electrical			
Overload	3.7.4	4.7.4)	1.5 dphu
Drum dial control	3.7.5	4.7.5)	for the
Input power	3.7.6	4.7.6)	group

* Defects per hundred units.

4.5.3 Group C inspection. This inspection shall consist of the inspection specified in Table III and shall be performed on sample units that have been subjected to and met Group A and B inspection. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for inspection.

4.5.3.1.1 Four (4) samples of complete analyzer shall be selected at random, one (1) sample for each Group 1 through Group 4 in Table III. The samples selected shall be at the start of the contract from the first quality conformance inspection lot. The samples shall constitute Group C requirement for first 200 units produced.

4.5.3.1.2 Thereafter, two (2) samples of complete analyzer shall be selected at random, one (1) sample for each Group 3 and Group 4 in Table III from every 200 units and fraction thereof produced.

4.5.3.1.3 Contracts for more than 100 and less than 200 units shall have Group C on first production lot as specified in 4.5.3.1.1 and another Group C on a production lot within the last 25 units produced as specified in 4.5.3.1.2.

4.5.3.2 Evaluation of samples submitted to individual Group C inspection. After referenced samples in 4.5.3.1 have been subjected to an individual Group C inspection test and prior to subjection to the next Group C test (unless otherwise specified in the individual test) the sample shall be subject to and meet operational requirement and test of 3.7.7 and 4.7.7.

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4.5.3.3 Noncompliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Group A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.3.4 Order of inspection within Group C. Group C inspection shall be performed in any order which is satisfactory to the Government.

TABLE III. Group C inspection

Inspection	Req. Para.	Test Para.
Group 1		
Fungus	3.8.1	4.8.1
Group 2		
Salt fog	3.8.2	4.8.2
Group 3		
Temperature	3.8.3	4.8.3
Humidity	3.8.4	4.8.4
Shock, drop	3.8.5	4.8.5
Group 4		
Altitude	3.8.6	4.8.6
Immersion	3.8.7	4.8.7
Vibration	3.8.8	4.8.8
Bounce	3.8.9	4.8.9
Shock, bench-handling	3.8.10	4.8.10

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4.5.4 Reinspection of conforming Group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are subjected to and pass Group A inspection.

4.5.5 Disposition of Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order provided units meet requirements of 4.5.4.

4.6 Bounce preconditioning. Each Analyzer ZM-3( )/U less Case CY-1121 ( )/U produced shall be placed in its normal operating position on the table of a package tester of suitable capacity as made by the L.A.B. Corporation, Skaneateles, New York, with accessories as per Model 1000SC and SC-A-46439, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the equipment under test to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute and shall meet the requirements of 3.12.

4.7 Electrical.

4.7.1 Accuracy.

4.7.1.1 Capacitance. Standard capacitors having certified known values to accuracies of ±1 percent shall be used to test the center, upper and lower extremes of each range of the analyzer. Set the selector switch to "CAPACITANCE." Connect the test capacitor to the capacitance terminals and set the range switch to the corresponding range. The accuracy of the indicated capacitance values shall meet the requirements of 3.7.1.1.

4.7.1.2 Insulation resistance. Standard resistors having certified known values to accuracies of ±1 percent shall be used to test the center and ends of each range of the analyzer. The accuracy of the readings obtained on the insulation resistance dial when tested as follows shall meet the requirements of 3.7.1.2.

(a) Set the selector switch to "INSULATION RESISTANCE" and the megohm switch to the "X 1" position. Select the proper resistor and connect to the insulation resistance terminals. Depress the key switch and adjust the dial control until the meter pointer remains constant whether the key switch is depressed, or not.

(b) Place the megohm switch in the "X 100" position and repeat procedure in (a) above.

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4.7.1.3 Power factor. Set the selector switch to "CAPACITANCE" and the range switch to "4" (.04 to 30 uF) and perform the following tests:

(a) Connect a 1 uF, ±1 percent capacitor, in series with a 318 ohm ±0.5 percent resistor, to the capacity terminals. Adjust the power factor and dial controls for maximum dip. The power factor factor dial shall show 20 ±2 and shall meet the requirements of 3.7.1.3.

(b) Connect a 1 uF ±1 percent capacitor in series with a 637 ohm ±0.5 percent resistor to the capacity terminals. Adjust the power factor and dial controls for maximum dip. The power factor dial shall show 40 ±4 and shall meet the requirements of 3.7.1.3.

4.7.1.4 Leakage. Set the selector switch to "LEAKAGE" and the voltages shall be measured, with a high impedance dc voltmeter having an accuracy of ±1 percent full scale, at points specified in the following tests:

(a) Short circuit current. With the operating voltage set for 500 volts on the voltmeter, the leakage terminals shall be shorted. Upon throwing the meter switch to the 50 ma position, the meter shall read between 35 and 45 ma, and shall meet the requirements of 3.7.1.4.

(b) Load current. A 10,000 ohm, 20 watt, 1 percent resistor shall be connected across the leakage terminals together with the dc voltmeter. The meter switch shall be placed at 50 ma and the operating voltage control shall be adjusted until the external meter reads 300 volts. The analyzer meter shall read 30 ma ±1.5 ma, and shall meet the requirements of 3.7.1.4.

(c) Voltmeter accuracy. Connect the dc voltmeter to the leakage terminals. Set the analyzer to 60 volts. Set the meter switch to the 50 ma position, and record the external voltmeter reading. Repeat for 300, and 600 volts. The readings shall meet the requirements of 3.7.1.4.

4.7.2 Voltage on capacitance terminals. A high impedance ac voltmeter with an accuracy of ±1 percent full scale shall be attached to the leakage terminals of the analyzer to measure the ac voltage present and the voltage readings shall meet the requirements of 3.7.2.

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4.7.3 Discharging circuits. Set the selector switch to "LEAKAGE". Connect a 10 uf 600 Vdc capacitor to the leakage terminals. Apply 600 volts by placing the meter switch in the 50 ma position. Release the meter switch. Remove the capacitor and place it across a high impedance voltmeter, having an accuracy of ±1 percent full scale, to determine if a charge remains on the capacitor. Results shall meet the requirements of 3.7.3.

4.7.4 Overload. A shorting wire shall be attached separately and at different times to the leakage, insulation resistance, or capacitance terminals. The analyzer shall be turned on for 5 minutes with the shorting wire in place. At the end of this time, the shorting wire shall be removed and the analyzer shall meet the requirements of 3.7.4.

4.7.5 Drum dial control. The drum dial control shall be rotated from one end to the other and back again 300 times and shall meet the requirements of 3.7.5.

4.7.6 Input power. The analyzer shall be connected to an ac power source with an accuracy of ±1 percent. The capacitor shall be attached to the capacitance terminals and a reading taken with input voltage of 115 Vac at 60 Hz. The input voltage shall be varied over the voltage and frequency range specified and shall meet the requirements of 3.7.6.

4.7.7 Operational. Each analyzer shall be tested with standard capacitors, having certified known values (to accuracies of ±1 percent) of capacitance, leakage, insulation resistance and power factor. The operation of the analyzer shall meet the requirements of 3.7.7.

4.8 Service conditions. The Analyzer ZM-3( )/U shall be tested in accordance with the following service conditions.

4.8.1 Fungus. The analyzer in its normal operating configuration shall be subjected to 28 days of the fungus test specified in Procedure I, Method 508 of MIL-STD-810 and shall meet the requirements of 3.8.1.

4.8.2 Salt fog. The analyzer in its normal operating configuration shall be exposed to salt fog in accordance with Procedure I, Method 509 of MIL-STD-810 for 48 hours and shall meet the requirements of 3.8.2.

4.8.3 Temperature.

4.8.3.1 Low temperature. The analyzer in its normal operating configuration shall be subjected to the low temperature test specified in Procedure I, Method 502 of MIL-STD-810, with temperatures in steps 2 and 4 held for 2 hours each, and shall meet the requirements of 3.8.3.1.

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4.8.3.2 High temperature. The analyzer in its normal operating configuration shall be subjected to the high temperature test specified in Procedure I, Method 501 of MIL-STD-810, with temperature in step 4 held for 2 hours, and shall meet the requirements of 3.8.3.2.

4.8.4 Humidity. The analyzer in its normal operating configuration shall be subjected to the humidity test specified in Procedure II, Method 507 of MIL-STD-810 and shall meet the requirements of 3.8.4.

4.8.5 Shock, drop. The analyzer in the transit case CY-1121( )/U shall be dropped a distance of 18 inches on each face, edge, and corner for a total of 26 drops. Drops shall be made from a quick-release hook; or pendulum tester as made by L.A.B. Corporation, Skaneateles, New York, or equal. The floor or barrier receiving the impact shall be of two inch fir backed by concrete or a rigid steel frame. The equipment shall meet the requirements of 3.8.5.

4.8.6 Altitude. The analyzer in its normal operating configuration shall be subjected to the altitude test specified in Procedure I, Method 500 of MIL-STD-810, with altitude in step 3 held for 1 hour, and shall meet the requirements of 3.8.6.

4.8.7 Immersion. The case CY-1121( )/U, less the unit, shall be braced (for example: dummy front panel and chassis) and weighted to simulate the weight of the equipment. The cover shall be closed and the case immersed to a minimum depth of 3 feet in fresh water at room temperature for 2 hours. Immediately prior to immersion, the temperature of the transit case shall be 40°F or more, above the temperature of the water. The tank in which the transit case is immersed shall be of sufficient capacity to maintain the water within $\pm 5^{\circ}$F of its initial temperature or the temperature of the water shall be maintained within those limits by other means. After completion of the 2 hour period of immersion, the transit case shall be removed from the water and wiped dry on exterior surfaces. When the transit case is opened, there shall be no evidence of leakage as required by 3.8.7.

4.8.8 Vibration. The analyzer in its normal operating configuration shall be subjected to the vibration test as follows:

(a) Secure the equipment, less dust cover and transit case, directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured.

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(b) The equipment shall be vibrated with the frequency varying between 10 and 55 Hz at a total excursion of 0.03 ± 0.006 inch, the frequency varying uniformly from 10 to 55 Hz and returning to 10 Hz in approximately 1 minute. Application of the vibration test specified above shall be in each of the following directions:

(1) Horizontal, parallel to the major horizontal axis of the equipment for a period of 15 minutes for the specified frequency range of 10 to 55 Hz.

(2) Horizontal, at right angles to the major horizontal axis of the equipment for a period of 15 minutes for the specified frequency range of 10 to 55 Hz.

(3) Vertically to the major horizontal axis of the equipment for a period of 15 minutes for the specified frequency range of 10 to 55 Hz.

(c) The equipment shall be inspected to ascertain that no fixed parts have become loose or damaged. The operational measurements per 3.7.7 and 4.7.7 shall be performed after completion of vibration test, and the equipment shall meet the requirements of 3.8.8.

4.8.9 Bounce. The analyzer in its transit case CY-1121( )/U shall be tested on a package tester of suitable capacity as made by the L.A.B. Corporation, Skaneateles, New York, with accessories as per Model 1000SC and SC-A-46439, or equal. The test shall be as follows:

(a) Secure the equipment to the 48 x 48 inch vehicular adapter plate. The adapter plate shall be free of cracks and broken welds, and all unused holes shall be plugged. No part of the mounted surface shall be less than seven inches from the edge of the adapter plate.

(b) Cover the tester bed with a panel of 1/2 inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at six inch intervals around all four edges and at three inch intervals in a six inch square in the center.

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(c) Place the adapter plate on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than three inches and not less than one inch. Additional barriers may be used to safeguard personnel, provided that the fore-and-aft motion of the adapter plate against the backstop is not restrained.

(d) Operate the package tester, shafts in phase, for a total of three hours at 284 RPM ± 2 RPM. Turn the equipment at the end of each 30 minutes so it will rest on a new surface.

(e) The equipment shall meet the requirements of 3.8.9.

4.8.10 Shock, bench-handling. The analyzer less the transit case and dust cover shall be positioned as for servicing on a solid two inch fir bench top and shall be subjected to the following:

(a) Tilt up the equipment through an angle of 30 degrees, using one edge of the equipment as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axis, for a total of four drops on horizontal face.

(b) Repeat (a) with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the equipment could be placed practicably during servicing.

(c) The equipment shall meet the requirements of 3.8.10.

4.9 Visual and mechanical. The analyzer shall be examined for the defects listed in MIL-STD-252 and shall meet the requirements of 3.9.

4.10 Interchangeability. The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirement of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of case, cover and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

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(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as case, covers, and mountings.

(d) Location of switches, sockets, and fasteners.

(e) Location of panel mounting screw holes on basic unit for mating with case.

4.11 Burn-in and debugging. Each analyzer produced shall be placed in its normal operating position on a test table with room temperature at 77°F ± 5°F, with specified operating voltage applied from a suitable power source. With all controls in the normal operating position, the equipment shall be turned on and remain on for a continuous 24 hour period. In case of parts failure, the equipment shall be repaired, recalibrated, and the burn-in and debugging test repeated and shall meet the requirements of 3.11. A failed item analysis report shall be submitted on each part failure with contents of report per 5.10.5 of MIL-STD-781.

4.12 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2(b)).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Analyzer ZM-3( )/U shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Analyzer ZM-3( )/U shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method 1C-1.

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5.1.1.4.2 Electron tubes. Remove all operating tubes over 4 inches in length not securely clamped in place and package with spare tubes, individually, in accordance with the applicable provisions of MIL-E-75.

5.1.1.4.3 Lamps. Each lamp shall be packaged individually in accordance with Method III of MIL-P-116. Cushion each lamp by wrapping in paperboard conforming to PPP-P-291, type 1 and secure with tape conforming to PPP-T-45, type 111.

5.1.1.4.4 Fuses. Fuses shall be packaged in accordance with Method III, of MIL-P-116. Cushion the fuses by wrapping together in paperboard conforming to PPP-P-291, type 1, and secure wrap with tape conforming to PPP-T-45, type 111.

5.1.1.4.5 Analyzer ZM-3( )/U Within Case, Test Set CY-1121/U. Each Analyzer ZM-3( )/U within Case, Test Set CY-1121/U shall be packaged in accordance with Method III of MIL-P-116. Wind the line cord into a coil of proportionate dimensions and tie at three equidistant places with twine conforming to T-T-871, type II, 6 ply, securing the plug to the coil with the ties. Place the tied coil between the side of the test set and the transit case. Close transit case cover and secure all fastenings.

5.1.1.4.6 Consolidated package. Tubes, lamps and fuses, packaged as specified in 5.1.1.4.2 through 5.1.1.4.4, shall be packed in a box conforming to PPP-B-636, type CF, class Domestic, variety SW, grade 200. Two dimensions of the box shall be equal to the width and depth dimensions of the transit case. Fill all voids within the box with paperboard cushioning conforming to PPP-P-291, type I. Close the box in accordance with the appendix of the box specification. The box, together with the transit case packaged as specified in 5.1.1.4.5, shall be overpacked within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Cushioning, in the form of cells or built-up pads, fabricated from fiberboard conforming to PPP-F-320, type CF, class WR, grade W5c, shall be positioned between the cleats of the transit case to provide a smooth surface between it and the inside surfaces of the box and the spare parts box as applicable. In addition, a pad fabricated of fiberboard conforming to PPP-F-320, type CF, class Domestic, variety DW, grade 350 and equal to the end dimension of the transit case, shall be placed between the spare parts box and the transit case. Close the box in accordance with the appendix of the box specification. In addition, seal all seams and joints with tape, not less than two inches wide, conforming to PPP-T-76.

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5.1.2 Level C. Each Analyzer ZM-3( )/U shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2(b)).

5.2.1 Level A.

5.2.1.1 Palletized load. A quantity of Analyzer ZM-3( )/U, packaged as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s. The load shall be "bonded" to the pallet by strapping.

5.2.1.2 Less than palletized load. A quantity of Analyzer ZM-3( )/U packaged as specified in 5.1, shall be packed in a close-fitting fiberboard box, conforming to MIL-B-43291 or PPP-B-640, class 2, style E. The gross weight of the boxes conforming to PPP-B-640 shall not exceed 200 pounds. Closure shall be in accordance with the appendix to the applicable box specification. The shipping containers shall be reinforced by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV, and PPP-S-760, type II respectively; selection of the material and application shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Each Analyzer ZM-3( )/U packaged as specified in 5.1, shall be packed as specified in 5.2.1.

5.2.3 Level C. Analyzer ZM-3( )/U packaged as specified in 5.1, shall be packed as specified in 5.2.1, except that the pallet, fiberboard boxes and fiberboard caps shall be of the type, size and kind commonly used for the purpose and shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation. Reinforcing of boxes, as specified in 5.2.1.2, will not be required. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES.

6.1 Intended use. The Analyzer ZM-3( )/U is an instrument designed for general use in determining the essential characteristics of capacitors within the range of 5 micromicrofarads to 10,000 microfarads.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment (Levels A, B or C) (see section 5).

(c) If rough handling and functional tests are required.

(d) Quantity of first articles required.

(e) Place of final inspection.

(f) Technical literature required.

(g) Quantity of tools and running spare parts required.

(h) Marking for shipment and shipping containers.

6.3 Nomenclature and type designation. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. The parentheses in the type designation will be deleted or replaced by a letter identifying the particular design for example: SG-12F/U or AN/URM-73C.

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (see 4.5.3.3)

6.5 Temperature conversion. Temperature conversion from fahrenheit to centigrade as listed in this specification are as follows:

160°F	=	71°C
125°F	=	51.5°C
77°F	=	25°C
40°F	=	4.44°C
-4°F	=	-20°C
-65°F	=	53.9°C

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6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodians:
Army - EL
Navy - AS
Air Force - 82

Review activities:
Army - MI
Navy - SH, MC
Air Force - 82

User activity:
Army - ME, AV

Preparity Activity
Army - EL

Project Number - 6625-0313

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-A-12636B Analyzer ZM-3()/U

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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